

Waves

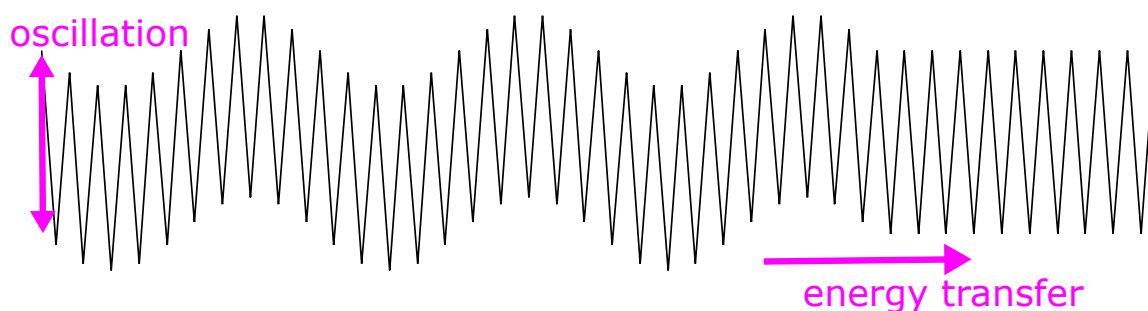
A wave carries _____ and can carry _____ without carrying _____ from one place to another.

Inside a wave, each part moves one way then the other. This is called _____. Each part's oscillation causes the next part to oscillate as well.

In a _____ wave the oscillations are in the _____ as the energy transfer.

In a _____ wave the oscillations are at _____ to the energy transfer.

- 1 Two friends stretch a long spring between them. One of them starts moving it up and down. After a short time, it looks like this:



(a) Is this a transverse or longitudinal wave?

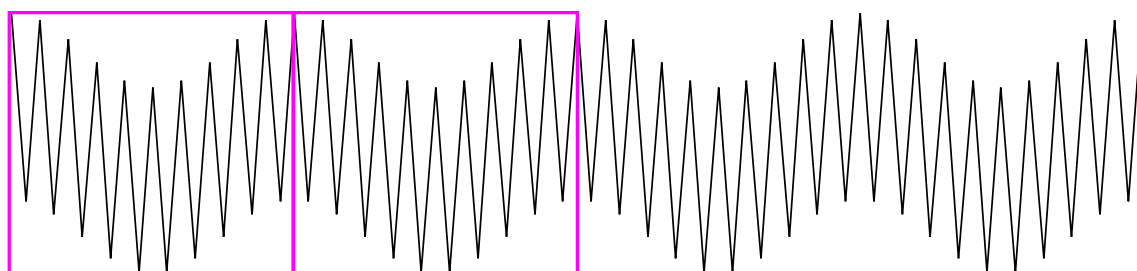
(b) The highest points on the wave are called _____. Label 3 peaks with the letter P.

(c) The lowest points are called _____. Label 3 troughs with the letter T.

- 2 Use a ruler to measure the distance between neighbouring peaks in question 1.

- 3 Use a ruler to measure the distance between neighbouring troughs in question 1.

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- 4 A short time later, the wave looks like this. Boxes have been drawn around sections of the wave.



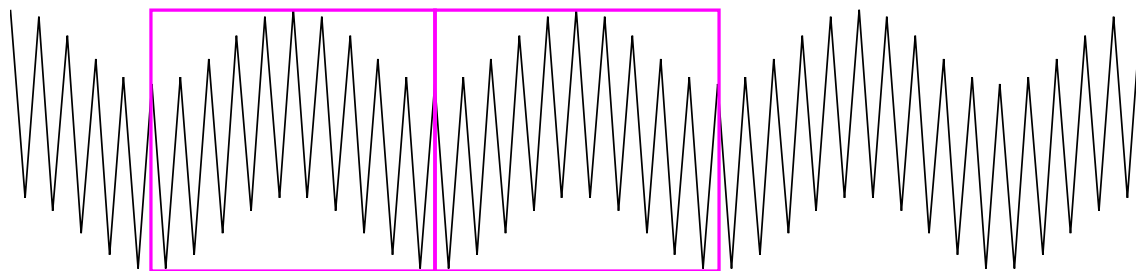
(a) Compare the contents of each box. Do they look the same?

(b) Each box contains a section of wave from one _____ to the next.

(c) Measure the length of each box.

(d) Draw the next box on the wave above.

5 The boxes can be drawn differently on the same wave.



(a) Each box contains a section of wave from one _____ to the next.

(b) Measure the length of each box.

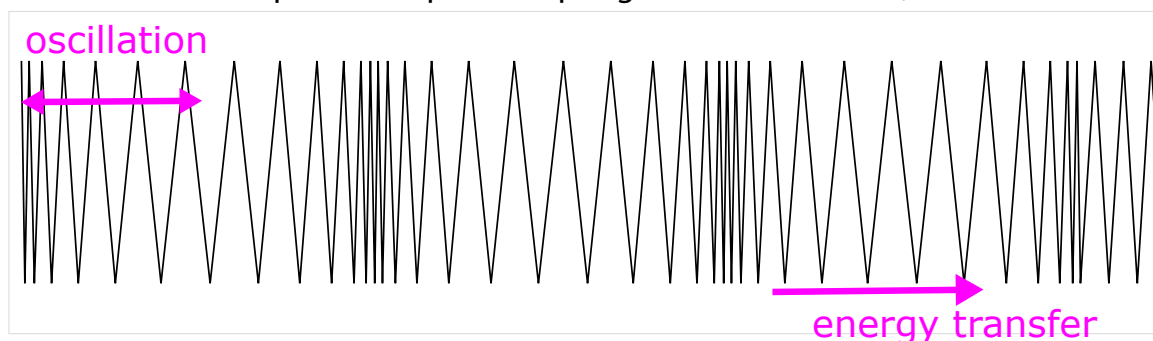
(c) Draw the next box on the wave above.

6 The boxes in questions 4 and 5 show a _____ in the wave. Their width is called the _____ of the wave.

(a) Write down the wavelength of the wave.

(b) Compare your answer to (a) with your answers to questions 2 and 3.

7 The two friends now push then pull the spring. After a short time, it looks like this:



(a) Is this a transverse or longitudinal wave?

(b) The places where the spring is bunched up on the wave are called _____. Label 3 compressions with the letter C.

(c) The places where the spring is most spaced out are called _____. Label 3 rarefactions with the letter R.

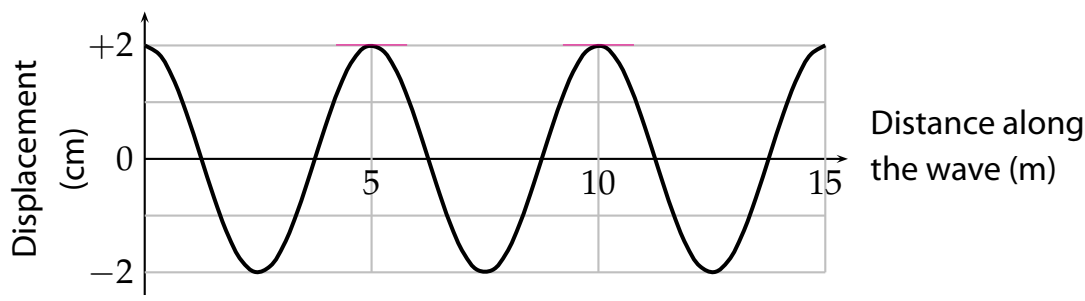
(d) Draw a box around a repeating unit on the wave.

8 Measure the wavelength of the wave in question 7.

_____ measures how far each part of the spring is from its resting place.

When we draw a graph of a wave, we plot

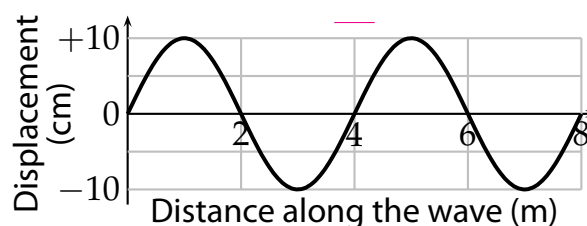
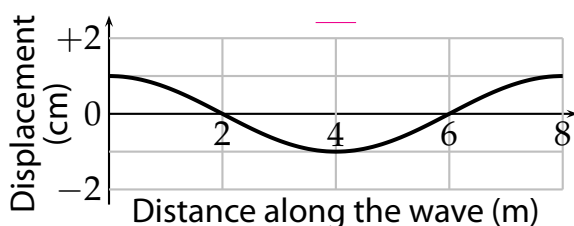
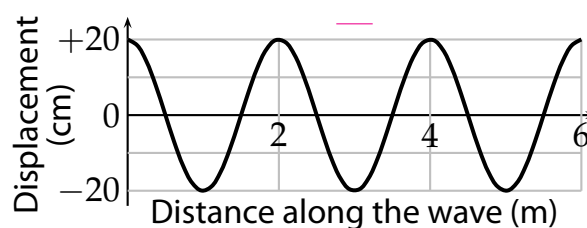
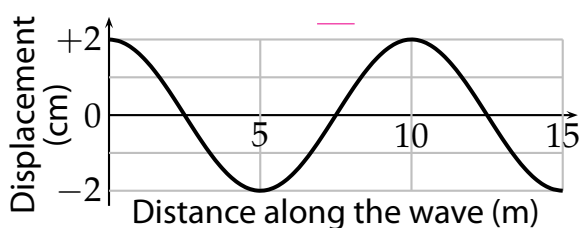
- the _____ on the vertical (y) axis, and
- the _____ on the horizontal (x) axis.



The _____ () is the largest displacement on the wave. Here, the amplitude is _____.

9 Using the scale on the graph, write down the wavelength of the wave above.

10 Write down the wavelength (λ) and amplitude (A) of each wave below.



Sound is an example of a _____ wave.

11 A citizen listens to the news on their TV. Complete the sentences using the words or phrases **towards**, **ears**, **brain**, **sound wave**, **speaker**, **oscillates**.

The _____ in the TV _____.

This makes air molecules move away from the speaker then _____ it.

The air carries the _____ to the listener.

When the wave reaches the listener, the air makes parts of their _____ oscillate.

Nerves detect this motion, and the information reaches the listener's _____.